

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

Mini Project Proposal (BCS-554)

Team Id: 26_CS_3J_02

Team Details:

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Project Title:

PrepWise – AI-Powered Mock Interview Platform
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Domain:

1. Software – Web Application
2. Artificial Intelligence / Machine Learning/ Deep Learning
3. Natural Language Processing
4. Big Data / Cloud Computing

Problem Statement:

Many students and job seekers struggle to prepare effectively for technical and HR interviews due to the lack of realistic interview practice and personalized feedback. Conventional interview preparation methods, such as reading questions or watching videos, do not provide an interactive experience or evaluate communication and problem-solving skills. Existing AI interview platforms are often expensive, generic, or lack role-specific customization. There is a need for an affordable, intelligent platform that simulates real interview scenarios, generates personalized questions based on job roles and skills, conducts interactive voice interviews, and provides instant AI-driven feedback. Such a solution can help users improve technical knowledge, communication skills, confidence, and interview readiness before appearing for actual recruitment processes.

Proposed Solution:

PrepWise is an AI-powered web application that simulates real-world technical and HR interviews through conversational voice interactions. Users can choose a job role and technology stack, after which the system generates personalized interview questions using Generative AI. The platform conducts interviews using AI voice agents, evaluates responses, and provides detailed feedback on communication, technical accuracy, confidence, and overall performance. Firebase Authentication and Firestore securely manage user accounts and interview history, allowing users to track their progress over time. The application features a responsive interface built with Next.js and Tailwind CSS, enabling users to practice interviews anytime from any device while receiving personalized recommendations for improvement.

Unique/Distinctive feature of the solution:

1. Uniqueness and Distinctive features:

PrepWise combines Generative AI, voice-based interviews, and automated performance analysis into a single platform. Unlike static question banks, it creates role-specific interview sessions dynamically and provides personalized feedback. Users can review previous interviews, identify weak areas, and continuously improve through AI-powered recommendations.

2. Differentiation from Competitors:

Unlike many commercial interview preparation platforms, PrepWise focuses on affordability, customization, and realistic voice interactions. It allows users to practice interviews for different technologies and job roles while storing interview history and performance analytics. The system integrates AI-generated questions, voice communication, and detailed feedback within one seamless application, making interview preparation more practical and accessible for students.

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Tools/Technology Uses:

1) Hardware Requirements:

- a) Processor: Intel Core i5 (8th Gen or above) / AMD Ryzen 5
- b) RAM: 8 GB (Minimum)
- c) Storage: 20 GB Free Space
- d) Graphics Card: Integrated Graphics (Dedicated GPU Optional)
- e) Others: Stable Internet Connection, Microphone, Speakers/Headphones

2) Software Requirements:

- a) Operating System: Windows 10/11, macOS, Linux
- b) Development Tools: Visual Studio Code, Node.js, Git, npm
- c) Database: Firebase Firestore
- d) Web Browser: Google Chrome / Microsoft Edge / Mozilla Firefox
- e) Others: Next.js, React, TypeScript, Tailwind CSS, Firebase Authentication, Google Gemini API, Vapi AI, Vercel (Deployment)

Expected Outcome:

- a) AI-generated interview questions based on selected job role and skills.
- b) Real-time voice-based mock interviews.
- c) Personalized AI feedback and performance analysis.
- d) Secure authentication and interview history management.
- e) Interactive dashboard to track interview progress.
- f) Improved interview confidence and job readiness for students.

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(Proposal Evaluation Sheet Should be printed on a Separate Page)

(To be Filled by Faculty/Evaluator)

Proposal Evaluation:

1. Right Identification of the Problem (Appropriate selection of the problem)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable
2. Relevance of the Solution (Adequately addressing the problem/need)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable
3. Innovativeness in the Solution (Distinctive innovative components/features of the solution)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable
4. Uniqueness of the Solution (Intellectual Property Component)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable

Improvements/ Suggestions by the Evaluator:

1.	
2.	
3.	
4.	

Name of Faculty In-charge:

Designation:

Signature with Date: